

SignTalk AI — Technical Requirements Document (TRD)

System architecture, ML pipeline, APIs, data, deployment, testing and security

SignTalk AI | B.Tech CSE (AI/ML) | GLA University

1. Reference Architecture

Camera → OpenCV → MediaPipe Hands/Pose/Face → landmark normalization/fusion → graph construction → ST-GCN → sequence representation → Transformer/NLP → text caption.

Application: React/TypeScript frontend → WebSocket/WebRTC or HTTP boundary → FastAPI inference service → AI pipeline. Future edge deployment can move inference to the device using ONNX Runtime/TensorRT or another suitable runtime.

2. Technology Stack

Layer	Technology / Option	Purpose
Frontend	React + TypeScript	UI/live captions
Video	OpenCV	Capture/preprocessing
Landmarks	MediaPipe Hands/Pose/Face	Structured visual features
ML	Python + PyTorch	Training/inference
Graph	GCN / ST-GCN	Spatial-temporal learning
Language	Transformer	Sequence modeling
NLP	Python NLP/model tooling	Normalization/sentence reconstruction
Backend	FastAPI	Inference/API
Realtime	WebSockets/WebRTC	Low-latency streaming
Optimization	ONNX Runtime/TensorRT	Edge acceleration
Packaging	Docker	Reproducible deployment
Testing	PyTest/Postman/Playwright	Unit/API/E2E

3. Computer Vision Pipeline

1. Capture frames.
2. Resize/normalize as required.
3. Detect hands and detailed landmarks.
4. Detect body pose/keypoints.
5. Extract facial/non-manual cues where enabled.
6. Synchronize features by timestamp.
7. Normalize coordinates.
8. Fuse features.
9. Create temporal windows.

4. Graph Representation

Nodes represent selected landmarks such as finger joints, wrists, elbows and body/facial points. Edges represent anatomical or task-relevant relationships. Temporal edges connect corresponding nodes across frames. The exact graph schema must be documented with the final model.

5. Model Requirements

Model	Input	Output	Purpose
Baseline	Landmark sequence	Gesture class	Reference performance
ST-GCN	Spatial-temporal landmark graph	Sign representation/class	Joint relationships + motion
Transformer	Sign sequence/representation	Text sequence	Sequence-to-text
NLP	Model output	Normalized sentence	Readability/grammar

6. Training Requirements

- Separate train/validation/test sets.
- Prefer signer-independent test data.
- Track dataset/preprocessing/model versions.
- Use justified augmentation and class balancing.
- Record seeds/configuration.
- Save checkpoints and metrics.
- Never tune on the test set.
- Run modality/architecture ablations.

7. Data Model

Entity	Important fields
User	id
Session	sessionId, language, mode, timestamps
Frame	timestamp, metadata
Pose	keypoints
Hand	landmarks
Face	features/landmarks
Gesture Sequence	ordered normalized keypoints
Translation Output	text, confidence, timestamp
Model	version, architecture, checksum
Evaluation	dataset version, metrics, configuration

8. API Requirements

Endpoint	Method	Purpose
/api/session/start	POST	Start session
/api/translate/frame	POST	Process frame/keypoints
/api/translate/stream	WebSocket	Continuous translation
/api/model/info	GET	Model metadata

/api/languages	GET	Supported languages
/api/session/end	POST	End session

9. Configuration & Secrets

- Use environment variables.
- Keep .env out of version control.
- Separate development/testing/production configuration.
- Never hard-code credentials.
- Do not log camera frames.
- Document model paths and versions.

10. Performance Engineering

Stage	Design target
Frame capture	<20 ms where hardware permits
Landmark extraction	<50 ms where hardware permits
ST-GCN	<100 ms where hardware permits
Transformer/NLP	<150 ms where hardware permits
UI/update	<50 ms where hardware permits
End-to-end	<500 ms design target; benchmark experimentally

These are engineering targets, not achieved results. Actual measurements must be collected on selected hardware.

- Profile pipeline stages.
- Reduce unnecessary frame processing.
- Quantize models.
- Use ONNX conversion where appropriate.
- Use TensorRT where supported.
- Consider pruning/distillation if accuracy remains acceptable.

11. Security & Privacy

Area	Requirement
Video	No raw-video retention by default
Transport	Secure transport for remote communication
Authentication	Protect private sessions/APIs
Authorization	Restrict administrative/model operations
Secrets	Environment variables or secret manager
Logs	Exclude raw video/sensitive content
Retention	Minimal documented retention
Datasets	Respect licenses/institutional policy

12. Testing

Level	Tools	Coverage
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Unit	PyTest	Preprocessing/normalization/utilities
Model	PyTorch/sklearn	Accuracy/precision/recall/F1
API	Postman/HTTP tests	Critical endpoints
E2E	Playwright	Core UI flows
Performance	Profiler	Latency/FPS/memory
Offline	Network isolation	Local inference
Security	Dependency scanners/API tests	Dependencies/auth/input handling

13. Robustness Matrix

Condition	Variants
Lighting	Bright / normal / low
Background	Plain / cluttered
Camera	Different distance/angle
Speed	Slow / normal / fast
Occlusion	Partial hand occlusion
Signer	Known / unseen where possible
Handedness	Left/right or mirrored cases where appropriate
Sequence	Short / medium / longer supported sequences

14. Deployment

Academic deployment can use a React frontend and FastAPI backend, packaged with Docker. The longer-term privacy-oriented design is local/edge inference so camera data need not be continuously uploaded. Cloud hosting, ONNX/TensorRT deployment, WebRTC and production infrastructure must only be described as implemented after actual build and testing.

15. Monitoring & Runbooks

Metric	Trigger	Action
Latency	Above target	Profile capture, landmarks, model and NLP
FPS	Below target	Reduce processing/model cost
Error rate	Above threshold	Inspect logs and reproduce
Memory	Excessive	Profile tensors/runtime
Accuracy	Below target	Inspect data/model quality

16. Risks & Mitigations

Risk	Mitigation
Insufficient dataset	Augmentation/transfer learning/additional licensed data
Hand occlusion	Multi-frame tracking/robust preprocessing
Latency	Profiling/quantization/ONNX/TensorRT
Regional variation	Broader signer/dialect data
Poor lighting	Preprocessing/user guidance
Translation errors	Better sequence modeling/confidence handling

Hardware limits	Lightweight models
Privacy	Local processing/minimal retention
Scope creep	Prioritized releases

17. Technical Definition of Done

- Landmark extraction works on test environment.
- Training/evaluation is reproducible.
- ST-GCN/baseline results are documented.
- Transformer/NLP output is evaluated on held-out data.
- Real-time inference is integrated.
- Latency/FPS are measured.
- Critical API/UI tests pass.
- Privacy behavior is verified.
- Unsupported vocabulary and limitations are documented.

18. Glossary

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
GCN	Graph Convolutional Network
ST-GCN	Spatial-Temporal Graph Convolutional Network
NLP	Natural Language Processing
ISL	Indian Sign Language
FPS	Frames Per Second
API	Application Programming Interface
ONNX	Open Neural Network Exchange
NFR	Non-Functional Requirement